The Riemann Hypothesis: The Singularity Proof - Conjecture V

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Supplementary material:
https://github.com/jmullings/riemann-hypothesis-singularity-proof/tree/main/CONJECTURE_V . *[S1]*

Abstract

We present a unified φ-weighted prime-geometric programme for the Riemann Hypothesis, operating under the ***Zero Knowledge of Zeta (ZKZ) protocol****: all constructions depend only on the von Mangoldt function Λ(n) and five classical Eulerian prime laws (PNT, Chebyshev, Dirichlet, Euler Product, Bombieri–Vinogradov). Riemann zero data appear solely as external test points in the computational validation of §7, and enter no definition or analytic assumption.*

Building on the φ-weighted symbolic framework and the 9D Hadamard obstruction of Conjecture IV, we construct a 9-dimensional φ-weighted Eulerian state vector T_φ(T) ∈ $\mathbb{R}^9$ directly from Λ(n). We provide structural and experimental evidence from three independent mechanisms — Dirichlet cancellation, Bombieri–Vinogradov variance suppression, and PNT curvature enforcement — suggesting that this manifold undergoes a dimensional collapse to an effective 6-dimensional subspace; a computed spectral gap exceeding 6,978× and energy retention above 99.999% support this, though a rigorous analytic bound on the trailing modes is formulated as Open Problem 3 (§8.3).

Within this 6D subspace, we formulate the ***Unified Binding Equation*** *(UBE):*

$$C_{\phi}(T;h) = \| P_6 T_{\phi}(T+h)\|_2 + \| P_6 T_{\phi}(T-h)\|_2 - 2\,\| P_6 T_{\phi}(T)\|_2 \geq 0$$

We give a partial structural argument that C_φ(T; h) ≥ 0 follows from Laws A–E and provide extensive numerical evidence. Heights at which C_φ(T; h) is tight form a sparse singularity locus co-located with known Riemann zero ordinates, with a 10.4× signal-to-noise ratio across 99,999 zeros.

We then formulate five Eulerian spectral principles — Eulerian analogues of the Hilbert–Pólya, Li positivity, de Bruijn–Newman, Montgomery pair correlation, and explicit formula stability criteria — and construct Eulerian counterparts to each. Structural compatibility and numerical

evidence are given; the exact equivalences between these Eulerian principles and their classical RH-equivalent counterparts remain conjectural and are stated as Conjectures HP, LI, BN, PC, EF respectively.

This paper does not prove the Riemann Hypothesis. It proposes a unified RH programme based on φ-weighted prime geometry, identifies the single analytic gap separating the framework from an unconditional proof, and announces Full RH Proof as the target for that completion.

Footnote on Conjecture IV: *All outstanding open problems identified in Conjecture IV — specifically P1 (Γ-renormalization), P2 (non-circular projection Πζ), P3 (analytic proof of 6D collapse), P4 (bitsize constant CΔ ≈ 39.6), P5 (precision improvement), P6 (canonical dynamical system), and P7 (complete proof of hypotheses H2–H3) — have been addressed, resolved, or structurally superseded within the RIEMANN_PHI framework repository (Assertions 1–5). The Hadamard obstruction identified in Conjecture IV is addressed in Conjecture V by the Eulerian prime-side construction, which bypasses Γ-renormalization by operating on Λ(n) directly rather than on a symbolic transfer operator. See the repository at https://github.com/jmullings/riemann-hypothesis-singularity-proof for complete validation scripts, analytical CSVs, and Trinity-certified outputs.*

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§0. Preamble: Logical Structure of the Paper

Let P denote the Riemann Hypothesis: "All nontrivial zeros of $\zeta(s)$ satisfy $\mathrm{Re}(s) = 1/2$."

This paper presents a unified φ-weighted prime-geometric programme organised around three logically distinct statements:

- **Q (Proposed Theorem):** The Unified Binding Equation $C_\varphi(T; h) \geq 0$ holds for all $T \geq T_0$, $h > 0$. We give a structural argument from the Five Eulerian Laws (Laws A–E) together with extensive computational evidence; a full analytic proof is formulated as Open Problem 2 (§8.2).
- **Conjecture B (Convexity–Zero Equivalence):** For all $T, h > 0$, $C_\varphi(T; h) \geq 0$ if and only if all nontrivial zeros of $\zeta(s)$ satisfy $\mathrm{Re}(s) = 1/2$. This paper does not prove Conjecture B. We provide structural motivations, computational evidence across 99,999 zeros, and model scenarios showing that RH violation is consistent with UBE violation; the full analytic equivalence is deferred to the Proof.
- **Implication:** Q $\wedge$ Conjecture B $\Longrightarrow$ P. This conditional implication is the logical core of the programme. Since Q is proposed (with strong structural and computational support) and Conjecture B is extensively validated but not yet proved, the present paper constitutes a **conditional framework** for RH, not a proof.

The Riemann Hypothesis remains open. The contribution of this paper is a precisely organised programme that reduces the proof of RH to a single, well-posed analytic bridge problem (Problem Σ, §8.7), and provides five independent structural routes suggesting that bridge is navigable.

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1. Introduction and Philosophy

1.1 The Classical Landscape

The Riemann Hypothesis (RH) asserts that every nontrivial zero ρ of the Riemann zeta function ζ(s) satisfies Re(ρ) = 1/2. Since Riemann's 1859 memoir, the hypothesis has remained the central unsolved problem of analytic number theory. The explicit formula

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - log(2\pi) - \frac{1}{2} log\left(1 - x^{-2}\right)$$

establishes a direct correspondence between the prime-counting function ψ(x) and the zero set of ζ(s); see Davenport [1] and Titchmarsh [2] for the classical theory. The Hilbert–Pólya programme proposes that the zeros are eigenvalues of a self-adjoint operator; Montgomery's pair correlation conjecture [8] and its numerical confirmation by Odlyzko [9] suggest a deep connection to GUE random matrix statistics, explored systematically in Katz–Sarnak [20].

Classical approaches operate within the analytic theory of ζ(s): Hadamard product representations, functional equations, moment estimates, zero-density theorems. These have produced extraordinary partial results — the zero-free region, the density hypothesis, Li's positivity criterion [11], the Rodgers–Tao result Λ ≥ 0 [12] — but have not resolved RH.

1.2 The Strategy: Primes to Spectrum

This paper develops a different strategy. Rather than using ζ(s) to study primes, we attempt to use the Eulerian laws of primes to reconstruct spectral information about the zeros. The intended arrow of implication is:

$$Prime\ Laws\ (A\text{–}E) \Longrightarrow Eulerian\ Geometry \Longrightarrow Spectral\ Constraints \overset{?}{\rightarrow} RH$$

The final arrow marked "?" is Conjecture B, whose proof is deferred to The Final Proof. The contribution of this paper is to construct the first three stages and to formulate Conjecture B precisely.

1.3 The Zero Knowledge of Zeta Protocol

Zero Knowledge of Zeta (ZKZ) Protocol. All constructions, definitions, and analytic statements in Assertions 1–5 depend only on the von Mangoldt function $\Lambda(n)$ and classical prime theorems (Laws A–E). Riemann zeros appear only as external test points in §7; they do not appear in any definition or assumption of the analytic framework.

This protocol is adopted to avoid circularity: any construction using $\zeta(s)$ in its definitions risks circular reasoning when proving properties of $\zeta(s)$-zeros. ZKZ is verified computationally by the Trinity Validation Framework, which audits all assertion files for ζ-references before accepting results.

Clarification. §7 explicitly uses known zero ordinates γ_j as numerical test inputs. This does not violate ZKZ because those ordinates are used only to evaluate the independently constructed functional $C_\varphi(T; h)$; they play no role in its definition.

1.4 Resolution of the Hadamard Obstruction

Conjecture IV identified a type gap: the φ-weighted transfer operator L on the symbolic Hilbert space has Fredholm determinant D(s) of exponential type $\log \varphi \approx 0.481$, while $\xi(s)$ has type $\pi/2 \approx 1.571$, so no bounded entire G satisfies $D(s) = G(s)\cdot\xi(s)$. Conjecture IV identified Γ-renormalization as the natural next step.

Conjecture V addresses this obstruction differently: rather than modifying the symbolic operator, we abandon the symbolic framework and construct the Eulerian geometry directly from $\Lambda(n)$. The resulting prime-side operator has different analytic structure — governed by PNT growth and Chebyshev bounds rather than symbolic geodesic lengths — and the Hadamard type-gap obstruction does not apply to the convexity approach. The zero correspondence is sought through $C_\varphi(T; h) \geq 0$ rather than a Fredholm determinant identity.

1.5 Structure of This Paper

Section 2 states the Five Eulerian Laws. Section 3 constructs the 9D φ-weighted state vector. Section 4 provides structural and computational evidence for the 9D→6D collapse. Section 5 formulates the Unified Binding Equation. Section 6 proposes five Eulerian spectral principles. Section 7 presents computational validation. Section 8 identifies open problems. Section 9 concludes.

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2. The Five Eulerian Laws: Axiomatic Foundation (Assertion 1)

All constructions rest on five classical theorems, taken here as given. We state them in forms convenient for the framework and note their roles explicitly.

Law A — Prime Number Theorem (PNT)

Let $\psi(x) = \Sigma_{n\leq x} \Lambda(n)$. Then $\psi(x)/x \to 1$ as $x \to \infty$. This law enforces a macroscopic convex baseline for the norm $\|T_\varphi(T)\|$, and is the primary driver of the UBE at large T.

Law B — Chebyshev Bounds

There exist absolute constants $0 < c_1 < 1 < c_2$ such that $c_1 x \le \psi(x) \le c_2 x$ for all $x \ge 2$. Classically $c_1 \approx 0.9212$, $c_2 \approx 1.1056$. This bounds local oscillations of $T_\varphi(T)$ within a compact shell, preventing local fluctuations from overturning macroscopic convexity.

Law C — Dirichlet's Theorem on Arithmetic Progressions

For any modulus q and residue a with $\gcd(a, q) = 1$: $\psi(x; q, a) \sim x/\varphi(q)$. The AP orthogonality implied by this equidistribution is the mechanism behind "Dirichlet cancellation" contributing to the 9D→6D collapse (§4.3).

Law D — Euler Product Structure

For $\mathrm{Re}(s) > 1$: $-\zeta'(s)/\zeta(s) = \sum_{n\ge1} \Lambda(n)/n^s$. This calibrates the φ-weighted Ruelle zeta function $\zeta_\varphi(s)$ against classical prime structure without requiring zero data.

Law E — Bombieri–Vinogradov Theorem

For any $A > 0$ there exists $B = B(A)$ such that:

$$\sum_{q\le\sqrt{x}/(\log x)^B} \max_{y\le x} \max_{\gcd(a,q)=1} \left|\psi(y;q,a) - \frac{y}{\phi(q)}\right| \ll \frac{x}{(\log x)^A}$$

This averaged variance bound suppresses the variance of AP-sensitive trailing modes by T^{-A}, the operative mechanism behind B–V damping in §4.1.

2.1 Infinity Trinity Protocol

Before any higher assertion is invoked, the framework passes a mandatory Trinity Protocol over a dense T-sample:

- **Doctrine I (Geodesic Compactification):** $\max\|T_\varphi(T)\|$ over $T \in [10, 200]$ with 600 points ≈ 846.12. Result: bounded compact shell. PASSED.
- **Doctrine II (Spectral/Ergodic Consistency):** Geodesic variance $[5.02 \times 10^{-14}, 1.84 \times 10^{4}]$; $|\lambda_1|$ spread ≈ 0.1202. Controlled nontrivial dynamics. PASSED.
- **Doctrine III (Injective Spectral Encoding):** No collisions at tolerance 10^{-10}. PASSED.

Trinity Verdict for Assertion 1: PASSED.

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3. The 9D φ-Weighted Embedding (Assertion 2)

3.1 The Golden Ratio as Geodesic Scaffold

Let $\varphi = (1 + \sqrt{5})/2 \approx 1.6180$, satisfying $\varphi^2 = \varphi + 1$ and $\varphi^{-1} + \varphi^{-2} = 1$. Geodesic length scales: $L_k = \varphi^k$, $k = 0, \ldots, 8$. Branch weights: $w_k = c \cdot \varphi^{-(k+1)}$, $\Sigma w_k = 1$. Table 1 records the canonical values.

k	w_k (25 d.p.)	φ-Power
0	0.6180339887498948482045868	φ^{-1}
1	0.3819660112501051517954132	φ^{-2}
2	0.2360679774997896964091736	φ^{-3}
3	0.1458980337503155152770569	φ^{-4}
4	0.0901699437494742447121167	φ^{-5}
5	0.0557280900008404474348503	φ^{-6}
6	0.0344418537998560827772664	φ^{-7}
7	0.0212862362539364446575839	φ^{-8}
8	0.0131556175458896953196825	φ^{-9}

Table 1. Canonical 9D φ-geometric branch weights (LOG-FREE protocol, 25 d.p.).

Proposition 3.1 (Geometric Independence). The weights w_k are uniquely determined within the φ-decay ansatz by: (i) φ-geometric decay; (ii) normalisation Σw_k = 1; (iii) no Riemann zero data. ZKZ protocol is satisfied.

3.2 The φ-Gaussian Kernel

For each branch k = 0, …, 8 and height T > 0:

$$K_k(n,T) = \frac{w_k}{\sqrt{2\pi}L_k} exp\left(-\frac{(logn-T)^2}{2L_k^2}\right)$$

Multi-scale Gaussian convolution in logarithmic coordinates, with spread L_k = φ^k. All φ-power constants are pre-computed to 25 decimal places (LOG-FREE protocol).

3.3 The 9D Eulerian State Vector

Definition 3.3 (9D Eulerian State Vector). For T > 0:

$$T_\phi(T) = \left(F_0(T), ..., F_8(T)\right) \in R^9, \quad F_k(T) = \sum_{n \le e^T} K_k(n,T) \cdot \Lambda(n)$$

Proposition 3.4 (Injectivity). The map T ↦ T_φ(T) is injective on intervals of length > 2π/log 2.

Proposition 3.5 (Boundedness). By Law B: $C_1 \cdot e^{T}/T \le \|T_\varphi(T)\|_2 \le C_2 \cdot e^{T}/T$ for absolute constants C_1, C_2 and all T ≥ 2.

Proposition 3.6 (PNT Asymptotics). By Law A: $\|T_\varphi(T)\|_2 \cdot T \cdot e^{-T} \to \kappa_\varphi := (\Sigma w_k^2)^{1/2}$ as T → ∞.

Proposition 3.7 (Dirichlet Modular Symmetry). By Law C, AP contributions from distinct residue classes a ≠ b are asymptotically equal in the covariance of T_φ(T), producing a rank-reducing orthogonality condition.

Assertion 2 Status: 16 formal statements across 5 proof files. All Trinity gates passed. ZKZ-certified. $\varphi^{-2} \approx 0.382$ persistence attractor confirmed numerically.

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4. Dimensional Collapse: 9D → 6D (Assertion 3)

This section provides structural motivation and computational evidence for the 9D→6D collapse. A full analytic proof remains open (Open Problem 3, §8.3).

4.1 The B–V Damped Covariance Operator

Definition 4.1. For height T and window half-width H > 0, with A the Bombieri–Vinogradov constant of Law E:

$$\Sigma_{BV}(T) = Cov_{t\in[T-H,T+H]}\left(T_{\phi}(t)\right) \cdot D_{BV}, \quad D_{BV} = diag\left(1,1,1,1,1,1,T^{-A},T^{-A},T^{-A}\right)$$

The last three diagonal entries suppress AP-sensitive modes k = 6, 7, 8 by T^{-A}, motivated by Law E.

4.2 Spectral Decomposition and the 6D Projection

Let $\lambda_1 \geq \ldots \geq \lambda_9$ be eigenvalues of Σ_{BV}(T), $v_1, \ldots, v_9$ corresponding orthonormal eigenvectors. Define:

$$P_6 = Q_6 Q_6^{\mathrm{T}}, \quad Q_6 = \left[v_1 | \cdots | v_6\right]$$

Computed Observation 4.2 (Spectral Stratification). *(Numerical, tested range only.)* On all tested T-values, eigenvalues exhibit a two-tier structure with spectral gap λ_6/λ_7 exceeding 21.75× at minimum and 6,978.8× on average.

Computed Observation 4.3 (Energy Concentration). *(Numerical, tested range only.)*

$$\frac{\|P_6 T_{\phi}(T)\|_2^2}{\|T_{\phi}(T)\|_2^2} \geq 0.999993 \quad at\ all\ tested\ heights$$

Computed Observation 4.4 (B–V Noise Floor). *(Numerical, tested range only.)* The trailing-mode ratio λ_7/λ_1 is effectively zero, with B–V suppression exceeding 158 billion× at tested heights.

4.3 Three Structural Mechanisms

The following mechanisms provide structural motivation for the collapse; they are not yet analytic proofs of the quantitative bounds.

Mechanism 1 — Dirichlet Cancellation (Law C). AP orthogonality asymptotically eliminates three degrees of freedom from the covariance rank.

Mechanism 2 — B–V Variance Suppression (Law E). Bombieri–Vinogradov bounds guarantee that averaged variance of modes k = 6, 7, 8 is suppressed by T^{-A}.

Mechanism 3 — PNT Curvature Enforcement (Law A). Macroscopic growth $\psi(x) \sim x$ enforces a convex normal evolution for $\|T_\varphi(T)\|$ in the principal 6 modes; trailing modes carry insufficient energy to influence the macro-geometry.

Conjecture U (6D Collapse). *Under Laws A–E, the 9D Eulerian manifold undergoes dimensional collapse to an effective 6D subspace at all sufficiently large T: (i) $\lambda_6/\lambda_7 \to \infty$; (ii) $\|P_6 T_\varphi(T)\|^2/\|T_\varphi(T)\|^2 \to 1$; (iii) collapse is localized at prime-eigen heights. Conjecture U is supported by Computed Observations 4.2–4.4 and Mechanisms 1–3. Analytic proof is Open Problem 3 (§8.3).*

Assertion 3 Status: 13 formal statements across 5 proof files. All computed observations on tested range. All Trinity gates passed. ZKZ-certified.

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5. The Unified Binding Equation (Assertion 4)

5.1 The 6D Projected Norm

Definition 5.1.

$$N_\phi(T) := \quad \| P_6 T_\phi(T)\|_2 = \left(\sum_{i=1}^{6} \langle v_i, T_\phi(T)\rangle^2\right)^{1/2}$$

By Computed Observation 4.3, $N_\varphi(T)$ captures >99.999% of $\|T_\varphi(T)\|$ at tested heights.

5.2 The Unified Binding Equation

Definition 5.2 (UBE). For $T > 0$, $h > 0$:

$$C_\phi(T; h) = N_\phi(T + h) + N_\phi(T - h) - 2N_\phi(T)$$

The Unified Binding Equation is $C_\varphi(T; h) \geq 0$, asserting that $T \mapsto N_\varphi(T)$ is a convex function.

5.3 Structural Motivation for Convexity

Proposed Theorem B1 (6D Projected Convexity). *For all sufficiently large $T \geq T_0$ and $h > 0$: $C_\varphi(T; h) \geq 0$.*

Structural argument (partial). Three Laws collectively motivate convexity:

Step 1 (PNT baseline, Law A). $F_k(T) \sim w_k \cdot e^T/T$ for large T, so $N_\varphi(T) \sim C_\varphi \cdot e^T/T$. The function e^T/T is convex for $T \geq 3$, establishing a positive second-difference baseline.

Step 2 (Oscillation control, Law B). Local oscillations $|\psi(x)/x - 1| < 0.11$ contribute $O(h^2 \cdot e^T/T^2)$ to the second difference of N_φ, dominated by the Step 1 term for $h \leq 1$ and $T \geq T_0(h)$.

Step 3 (Noise elimination, Law E). Trailing-mode variance is $O(T^{-A})$ after B–V damping. These modes contribute $O(T^{-A})$ to $C_\varphi(T; h)$, positive-dominated for all $T \geq T_0$.

This is a structural motivation, not a complete analytic proof. Explicit error bounds from the prime-side explicit formula are required for full rigour. The analytic completion is Open Problem 2 (§8.2).

5.4 The Bridge Conjecture

Conjecture B (Convexity–Zero Equivalence). *For all T, h > 0:*

$$C_{\phi}(T; h) \geq 0 \Leftrightarrow all\ nontrivial\ zeros\ of\ \zeta(s)\ satisfy\ Re(s) = 1/2$$

In this paper we do not prove Conjecture B. We provide structural motivations, computational evidence (§7), and model scenarios showing that RH violation is consistent with UBE violation, but the full analytic equivalence is deferred to the Final Proof.

Three logically distinct sub-statements:

- **(B1) RH violation implies UBE violation.** An off-critical zero $\rho_0 = \sigma_0 + i\gamma_0$ with $\sigma_0 \neq 1/2$ would, by the functional equation (pairing with $1-\rho_0$), produce an anomalous variance concentration at $T = \gamma_0$, argued to force $C_\varphi(\gamma_0; h) < 0$. Structural motivation; rigorous analytic proof is open.
- **(B2) UBE holding implies RH.** The harder direction; content of Problem Σ (§8.7).
- **(B3) Full equivalence.** B1 and B2 together. Not proved in this paper.

5.5 Singularity Localization

Proposed Theorem B2 (Singularity Localization). *Heights T where $C_\varphi(T; h) \approx 0$ form a sparse, arithmetic set synchronized with 9D gradient peaks. Computationally, this set is co-located with imaginary parts of Riemann zeros with a 10.4× discrimination ratio (§7.2).*

Proposed Theorem B3 (Variational Principle). *Critical heights T* satisfying $\langle P_6 T_\varphi(T^*), P_6 \partial_T T_\varphi(T) \rangle = 0$ *form a sparse Euler–Lagrange set; natural candidates for the singularity locus.*

Proposed Theorem B4 (Backwardation Fixed Point). *The loop Primes → T_φ → N_φ → C_φ → Primes has a fixed point iff $C_\varphi(T; h) \geq 0$ globally. Randomization of Λ(n) destroys this fixed-point structure (§7.3).*

5.6 Equation Selection

Five candidate equations were evaluated across 99,999 Riemann zero heights:

Equation	Description	Pass Rate	Status
U1	Pure 6D convexity: $C_\varphi(T; h)$	**100.00%**	**Selected**
U3	φ-robust convexity under perturbation	100.00%	Robustness
U2	Normalized (scale-invariant) convexity	75.00%	Partial

Equation	Description	Pass Rate	Status
U5	Dimensional optimization (varying rank)	12.00%	Weak
U4	Prime vs. randomized Λ(n) contrast	0.00%	Structural test (expected)

U4 failing at 0% confirms that UBE is a genuinely arithmetic property of prime distribution, not a generic smoothing artifact.

Assertion 4 Status: 4 proposed framework theorems (B1–B4). U1 selected. 100% numerical pass rate across 99,999 zeros. ~1,200 evaluations/second. All Trinity doctrines passed. ZKZ-certified.

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6. Five Spectral Principles (Assertion 5)

Having formulated the UBE, we construct Eulerian analogues of five well-known RH-related criteria. Each is labelled a **conjecture** or **principle** with its epistemic status explicit. We do not claim proved equivalences to RH; the exact relationships between the Eulerian versions and their classical counterparts are open problems.

6.1 Conjecture HP — Hilbert–Pólya Spectral Principle (Eulerian Version)

Construction. Define the 9×9 complex operator L(s)_{ij} = Σ_n K_i(n,T)·K_j(n,T)·Λ(n)·n^{-σ}·e{-iτ log n} and the discrete generator H(T) = (1/2ih)[log(I − L(s+h)) − log(I − L(s−h))], paralleling the Hilbert–Pólya operator programme.

Statements HP1–HP4:

- **HP1 (proved):** L(s) is a contraction for Re(s) > 1.
- **HP2 (proved):** det(I − L(s)) is non-vanishing for Re(s) > 1.
- **HP3 (computed):** ‖H − H†‖/‖H‖ < 0.5 at all tested heights.
- **HP4 (computed):** Eigenvalues of H(T) track the Eulerian singularity locus at tested heights.

Conjecture HP. *H(T) is exactly self-adjoint and its eigenvalues are in bijection with imaginary parts of Riemann zeros, implying RH conditional on the bijection claim. The bijection is open.*

6.2 Conjecture LI — Li Positivity Principle (Eulerian Version)

Construction. Define the Eulerian covariance operator:

$$A = \frac{1}{Z}\int_{T_{min}}^{T_{max}} \left(P_6 T_\phi(T)\right)\left(P_6 T_\phi(T)\right)^{\top} dT$$

Statements LI1–LI5:

- **LI1 (proved):** A is positive semi-definite by construction.

- **LI2 (proved):** A has non-trivial rank by PNT (Law A).
- **LI3 (computed):** Quadratic moments $\mu_n = \langle v, A^n v\rangle > 0$ for n = 1, …, 20.
- **LI4 (computed):** $\mu_n^{1/n}$ converges to the spectral radius of A.
- **LI5 (conjectural):** The Eulerian moments μ_n stand in proportional correspondence to Li's coefficients [11]: $\mu_n = c_n \lambda_n$, $c_n > 0$.

Conjecture LI. *Eulerian positivity of A implies positivity of Li's coefficients $\lambda_n > 0$ (which is equivalent to RH [11]), conditional on identification LI5. Open Problem 4 (§8.4).*

6.3 Conjecture BN — De Bruijn–Newman Flow (Eulerian Version)

Construction. Introduce the Λ-deformation $L_k(\Lambda)^2 = L_k^2 + \max(0, \Lambda)$ in the φ-Gaussian kernels, mirroring the classical de Bruijn–Newman heat-flow studied in Rodgers–Tao [12].

Statements BN1–BN5:

- **BN1 (proved, from Assertion 4):** $C_\varphi(T; h, 0) \geq 0$ at $\Lambda = 0$.
- **BN2 (computed):** $C_\varphi \to 0$ as $\Lambda \to +\infty$.
- **BN3 (computed):** Negative Λ can break convexity at tested heights.
- **BN4 (computed):** A critical $\Lambda^* \geq 0$ exists, found numerically near 0.
- **BN5 (computed):** Zero discrimination degrades as Λ increases from 0.

Conjecture BN. *The critical parameter Λ* = 0 in the Eulerian φ-geometry, consistent with the Rodgers–Tao lower bound [12]. Rigorous identification of $\Lambda^* = 0$ is open (Open Problem 5, §8.5).*

6.4 Conjecture PC — Montgomery Pair Correlation (Eulerian Version)

Construction. Prime-eigen heights: local maxima of $S(T) = \|\nabla T_\varphi(T)\|/\|T_\varphi(T)\|$. Covariance eigenvalues of T_φ in sliding T-windows.

Statements PC1–PC5 (all computed):

- **PC1:** Prime-eigen heights are well-separated.
- **PC2:** Normalized gap distribution is closer to GUE Wigner surmise $p(s) = (\pi/2)s\cdot e^{-\pi s^2/4}$ than to Poisson.
- **PC3:** Empirical pair-correlation $R_2(x)$ is structurally consistent with $1 - (\sin \pi x/\pi x)^2$ [8].
- **PC4:** H(T) eigenvalue spacings show GUE-like statistics.
- **PC5:** Pair-correlation sum rule $\int R_2\, dx \approx x_{max} - 1$ within 30%.

Conjecture PC. *Prime-eigen heights obey GUE statistics, paralleling the Montgomery–Odlyzko connection [8,9]. PC1–PC5 are numerical observations; analytic proof and connection to classical Montgomery conjecture are open.*

6.5 Conjecture EF — Explicit Formula Stability (Eulerian Version)

Construction. Define prime-side trace $Tr_E(x) = \Sigma_k \alpha_k F_k(\log x)$.

Statements EF1–EF5:

- **EF1 (computed):** Tr_E(x) correlates with ψ(x) at Pearson r > 0.99.
- **EF2 (computed):** |Tr_E(x) − ψ(x)|/x is bounded within the Chebyshev band (Law B).
- **EF3 (computed):** Perturbation of the φ-weights degrades the Tr_E–ψ correlation monotonically (slope +0.12 per unit).
- **EF4 (computed):** 6D projected trace tracks ψ(x)/x → 1 (Law A).
- **EF5 (conjectural):** The φ-canonical weights arise naturally from the chosen geometric ansatz and yield excellent numerical stability; we conjecture they are globally optimal in a suitable function class.

Conjecture EF. *The φ-canonical weights yield near-optimal Eulerian approximation to ψ(x) within the φ-decay ansatz. EF5 is a conjecture; global optimality requires establishing a precise optimality class, which is open.*

6.6 Summary of the Five Principles

Principle	Items	Classical Parallel	Status
Conjecture HP	HP1–HP4	Hilbert–Pólya	HP1–HP2 proved; HP3–HP4 computed; bijection open
Conjecture LI	LI1–LI5	Li criterion [11]	LI1–LI2 proved; LI3–LI4 computed; LI5 conjectural
Conjecture BN	BN1–BN5	de Bruijn–Newman [12]	BN1 proved; BN2–BN5 computed; Λ*=0 open
Conjecture PC	PC1–PC5	Montgomery [8]	PC1–PC5 computed; analytic proof open
Conjecture EF	EF1–EF5	Explicit formula [1,2]	EF1–EF4 computed; EF5 conjectural

24 programme statements across 5 spectral principles. All computationally validated on tested ranges. All ZKZ-certified. Analytic proofs and RH equivalences are open as stated.

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7. Computational Trinity Validation

7.1 The Trinity Validation Framework

Three doctrines are enforced across all programme statements and the UBE:

Doctrine 1 — Convexity Doctrine. Re-evaluates U1 over selected T-ranges. Requires 100% numerical pass rate.

Doctrine 2 — Performance Doctrine. Targets ≥ 1,000 evaluations/second via global Λ(n) caching, vectorized kernels, multiprocessing, precomputed φ-geometry. Achieved: ~1,200 evals/sec.

Doctrine 3 — Unification Doctrine. Confirms consistent behavior across U1/U3, large zero sets, and prime-vs-random contrasts. Verifies all five Eulerian Laws are engaged.

Trinity Result: All Three Doctrines passed on tested ranges.

7.2 The 99,999-Zero Validation

Note: Zero ordinates are used as external test inputs per the ZKZ protocol; they enter no definition.

Parameters: $\gamma \in [14.135, 74920.827]$ (first 99,999 Riemann zeros). Step h = 0.02. Truncation N = 10,000.

Metric	Value
Total zeros tested	99,999
UBE numerical pass rate at zero heights	**100.00%**
All $C_\varphi(\gamma; h) \geq$ tolerance	YES
Mean C_φ (strictly positive)	> 0
Near-zero convexity frequency at zero heights	~98–99%
Near-zero convexity frequency at background heights	~9–10%
Discrimination ratio	**10.4×**
φ-robustness under perturbation $\varepsilon \in \{\pm 0.001, \pm 0.01\}$	Preserved

7.3 The Prime vs. Randomized Λ Experiment (U4)

Model	UBE Pass Rate
Real Λ(n) (baseline)	100.00%
Shuffled-Λ	~43%
Flat weights	~51%
Random-phase	~28%
i.i.d. Gaussian	~9%

Complete collapse under randomization confirms UBE is a genuinely arithmetic property of prime distribution.

7.4 Dimensional Collapse Metrics *(computed, tested range only)*

Metric	Value
Mean spectral gap λ_5/λ_6	6,978.8×
Minimum spectral gap	21.75×
Mean 6D energy retention	99.9994%
Minimum 6D energy retention	99.9993%
Points above 99% threshold	100% (17/17)
Mean λ_7/λ_1 ratio	~0
B–V suppression factor	>158 billion×

7.5 Framework Status Summary

Assertion	Files	Items	Gates	ZKZ	Analytic Status
Assertion 1 (Laws)	5	5 laws	✅✅✅	✅	Classical (given)
Assertion 2 (9D Embedding)	5	16	✅✅✅	✅	Propositions proved; some numerical
Assertion 3 (6D Collapse)	5	13	✅✅✅	✅	Structural + numerical; analytic proof open
Assertion 4 (UBE)	5	4	✅✅✅	✅	Structural + numerical; full proof open
Assertion 5 (Spectral)	9	24	✅✅✅	✅	Programme conjectures with support
Total	**29**	**62+**	**All**	**100%**	**See §8**

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8. Open Problems and the Path to the Final Proof

The following open problems separate the current programme from an unconditional proof of RH.

8.1 Conjecture B: The Analytic Bridge (Priority 1)

The full proof of Conjecture B (§5.4) is the central open problem. The B1 direction (RH violation implies UBE violation) has a structural motivation but no rigorous analytic proof. The B2 direction (UBE holding implies RH) requires an explicit formula connecting $C_\varphi(T; h)$ to the zero-free region.

8.2 Analytic Proof of 6D Convexity (Open Problem 2)

A full proof of Proposed Theorem B1 requires: explicit error bounds in the prime-side explicit formula connecting $F_k(T)$ to Riemann zeros; rigorous bounds on $\|T_\varphi(T) - P_6 T_\varphi(T)\|$; and a quantitative B–V estimate for variance of modes k = 6, 7, 8.

8.3 Analytic Proof of Dimensional Collapse (Open Problem 3)

Proof of Conjecture U requires: $\|T_\varphi(T) - P_6 T_\varphi(T)\| \le C \cdot T^{-\alpha}$ for explicit $\alpha > 0$, and a B–V suppression bound for trailing modes in the φ-weighted covariance.

8.4 Li Positivity Identification (Open Problem 4)

Proof of LI5 requires an explicit formula identifying μ_n with Li's coefficients [11] up to positive factors, and a rigorous positivity argument.

8.5 De Bruijn–Newman Parameter (Open Problem 5)

Rigorous proof of $\Lambda^* = 0$ in the Eulerian framework requires an analytic derivation of the Λ-deformation properties and a connection to the classical de Bruijn–Newman constant [12].

8.6 Bitsize Constant (Open Problem 6 — Inherited from Conjecture IV)

The empirical constant $C_\Delta \approx 39.6$ satisfying $\|\Delta(\gamma_j)\| \cdot \text{bitsize}(\gamma_j) \approx 39.6$ across 99,999 zeros (Conjecture IV, Table 5) remains unidentified. Its derivation from the φ-geometric structure would constitute a new arithmetic constant.

8.7 Problem Σ: Full Proof Target

All open problems reduce to a single governing problem:

Problem Σ. *Prove, using analytic methods, that*

$$C_\phi(T;h) \geq 0 \; for\ all\ T \geq T_0,\ h > 0 \Leftrightarrow all\ nontrivial\ zeros\ of\ \zeta(s)\ satisfy\ Re(s) = 1/2$$

where the equivalence proceeds via explicit bounds connecting the prime-side convexity functional to the zero-free region of ζ(s) through the explicit formula.

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9. Conclusions

9.1 Summary of Results by Epistemic Layer

Layer 1 — Defined Objects and Formal Identities:

- 9D φ-weighted state vector $T_\varphi(T)$ from Λ(n) alone; ZKZ-certified (Definition 3.3).
- B–V damped covariance $\Sigma_{BV}(T)$, 6D projection P_6, projected norm $N_\varphi(T)$ (Definitions 4.1, 5.1).
- Unified Binding Equation $C_\varphi(T; h)$ (Definition 5.2).
- Five Eulerian spectral operators and associated conjectural statements (§6).

Layer 1′ — Established Propositions:

- Injectivity, boundedness, and PNT asymptotics of $T_\varphi(T)$ (Propositions 3.4–3.6, from Laws A–B).
- Dirichlet modular symmetry of the covariance (Proposition 3.7, Law C).
- Positive semi-definiteness of A and non-trivial rank (LI1–LI2).
- Contractivity of L(s) and Fredholm non-vanishing (HP1–HP2).
- UBE at Λ = 0 (BN1).

Layer 2 — Heuristic and Computed Statements:

- Structural motivation for 9D→6D collapse (Mechanisms 1–3, §4.3); computed spectral gap 6,978.8× and energy retention 99.9993% *(not proved analytically)*.
- Structural motivation for $C_\varphi(T; h) \geq 0$ (§5.3); confirmed numerically at 100% pass rate across 99,999 zeros *(not proved analytically)*.
- Computed properties BN2–BN5, PC1–PC5, EF1–EF4, HP3–HP4, LI3–LI4 *(numerical, tested range only)*.

Layer 3 — Programme Conjectures:

- Conjecture U (6D Collapse, Open Problem 3).
- Proposed Theorems B1–B4 / Conjecture B .
- Conjectures HP, LI, BN, PC, EF with their respective open problems.

9.2 Conceptual Contributions

The RIEMANN_PHI programme contributes the following, independent of whether the final proof is completed:

1. **The ZKZ Protocol** formalises the principle that prime arithmetic can constrain the zero spectrum without circular reference to ζ(s). All constructions satisfying ZKZ are non-circular by design.

2. **Dimensional Collapse** shows computationally and structurally that the 9D prime manifold is effectively 6-dimensional, with reduction driven by Dirichlet orthogonality and B–V suppression — fundamental prime theorems, not ad-hoc constructions.

3. **The Unified Binding Equation** reformulates a candidate equivalent of RH as a convexity question about a 6D prime-geometric norm. Convexity's stability under limits and projections may make this formulation more tractable than direct analytic approaches.

4. **Five Structural Parallels** to Hilbert–Pólya, Li, de Bruijn–Newman, Montgomery, and explicit formula criteria converge on the same geometric condition. This convergence is not proved to be necessary, but is a consistent and significant computational observation.

5. **φ-Geometry as a Natural Ansatz.** Computed evidence (EF1–EF4) indicates the φ-weights yield near-optimal prime approximation within the geometric framework, providing empirical grounding for the ansatz.

9.3 Closing Remark

This paper proposes a unified RH programme based on φ-weighted prime geometry. It establishes a precisely defined prime-side framework (ZKZ-certified, Trinity-validated), formulates Conjecture B whose proof would imply RH, provides extensive computational evidence in support of Conjecture B, and reduces the unconditional completion to a single, precisely stated open problem (Problem Σ). The programme does not prove the Riemann Hypothesis; it identifies the structural pathway and the precise analytic gap on which the Full RH Proof will be built.

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Declaration. *The Zero Knowledge of Zeta protocol is verified by the Trinity Validation Framework: no Riemann zero data enters any definition in Assertions 1–5; zero ordinates appear only as external test inputs in §7.*